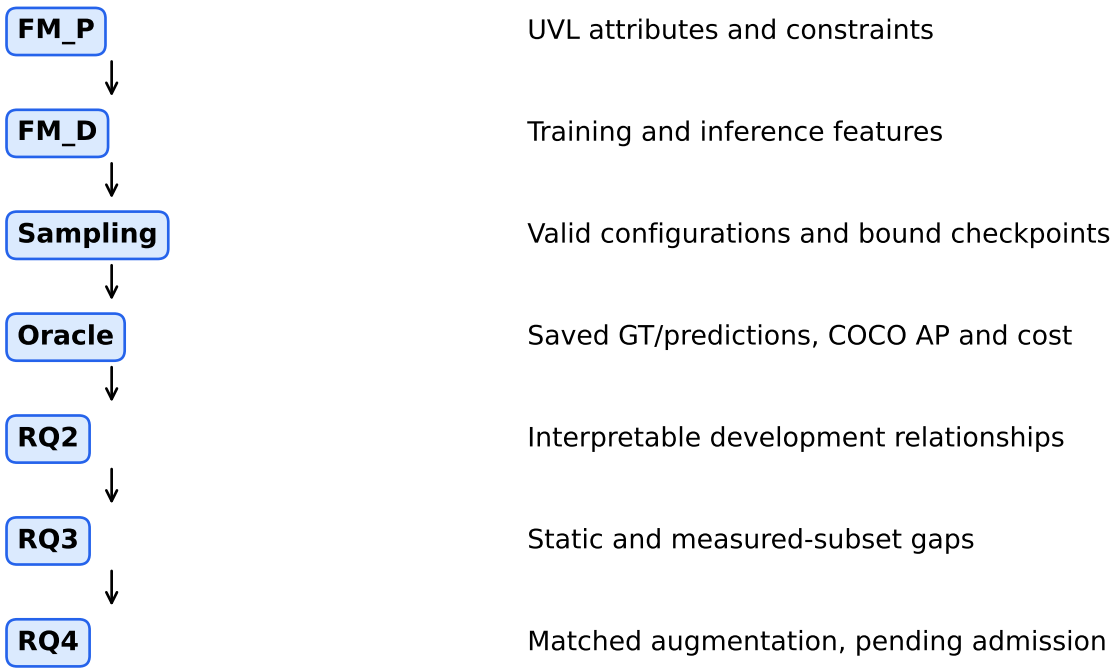
Dual variability research workflow



CONCEPTUAL_FRAMEWORK; completion tracked separately | Units: workflow steps

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: rq1_validation.json SHA256 dd7a8dcf4c31; full provenance in figure_evidence.json

Original UVL feature trees

FM_P

```
Degradation
    mandatory
        Illumination
            alternative
                Normal
                Overexposed_mild
                Overexposed_strong
                Underexposed_mild
                Underexposed_strong
                LowLight
    optional
        Blur
            or
                MotionBlur
                    alternative
                        Motion_k7
                        Motion_k15
                        Motion_k21
                DefocusBlur
                    alternative
                        Defocus_r3
                        Defocus_r6
        Noise
            or
                GaussianNoise
                    alternative
                        Gauss_s5
                        Gauss_s15
                        Gauss_s30
                ShotNoise
                    alternative
                        Shot_low
                        Shot_high
        Compression
            alternative
                JPEG_q75
                JPEG_q50
                JPEG_q25
        Atmosphere
            alternative
                Haze_light
                Haze_dense
                Rain
        Altitude
            alternative
                Downscale_x1_5
                Downscale_x2
        RollingShutter
```

FM_D

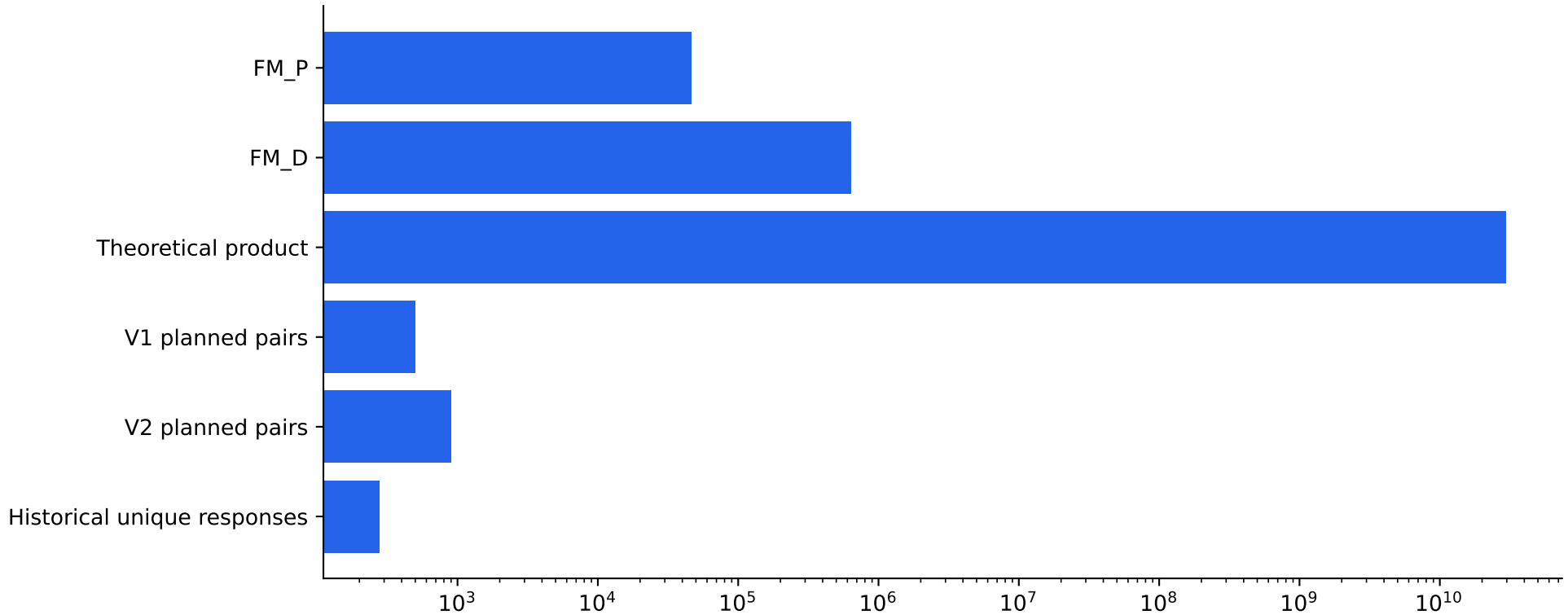
```
Detector
    mandatory
        Architecture
            alternative
                YOLOv8
                YOLO11
                YOLO26
                LEAF_YOLO
        ModelScale
            alternative
                Nano
                Small
        Weights
            alternative
                COCO_ZeroShot
                VisDrone_FT
        TrainResolution
            alternative
                Train640
                Train1280
        TrainAugmentation
            mandatory
                HSV_Default
                Geometric_Default
            optional
                Albu_Default
                NoiseAug
                MotionBlurAug
                ExposureAug
                JPEGAug
                HazeAug
        InferResolution
            alternative
                Infer640
                Infer960
                Infer1280
                Infer1536
        InferenceStrategy
            alternative
                FullFrame
                SAHI_Slicing
        PostProcessing
            alternative
                NMS
                EndToEnd_NMSFree
        Precision
            alternative
                FP32
                FP16
                INT8
    optional
        Preprocessing
            or
                CLAHE_Pre
                Gamma_Pre
                Denoise_Pre
                Deblur_Pre
        TTA
```

VALIDATED_LOGIC; not universal implementation support | Units: 25 / 34 leaves

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: fm_p.uvl SHA256 2033bb61456d; full provenance in figure_evidence.json

Configuration space and observed scope

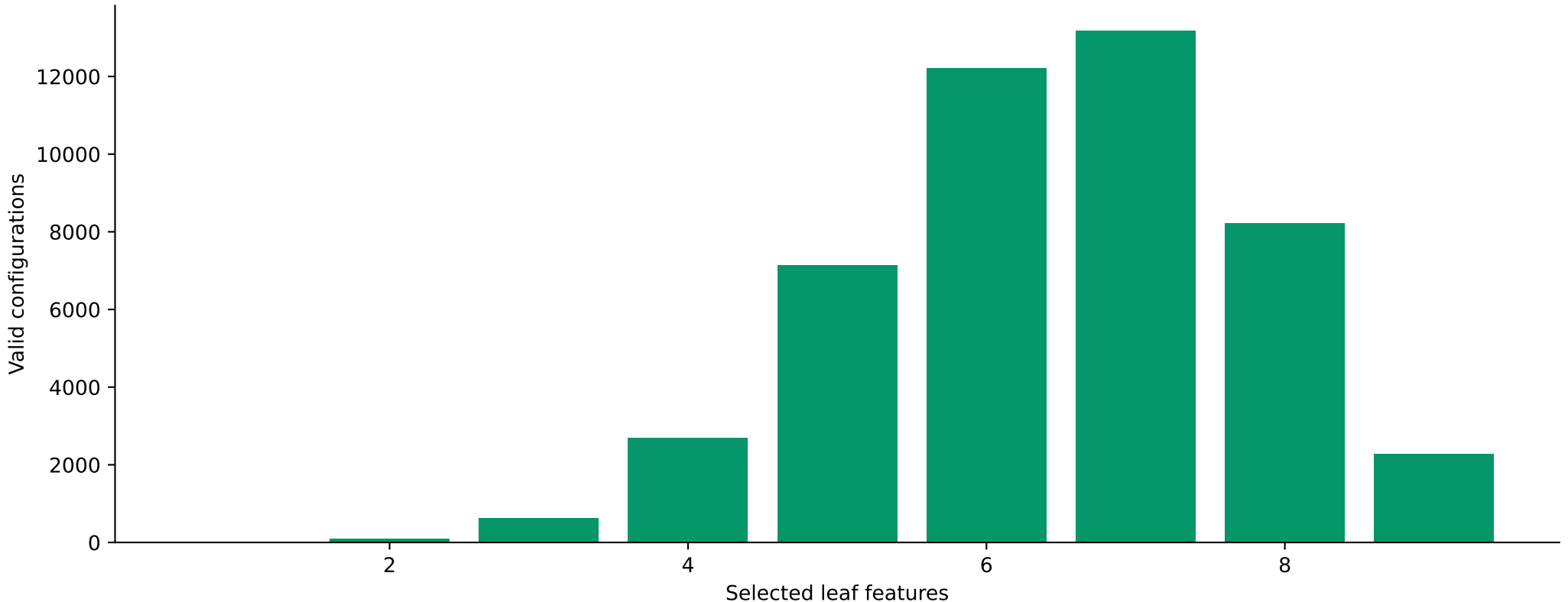


LOGICAL_COUNTS_AND_PLANS; historical responses are development only | Units: configuration count, logarithmic axis

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: rq1_validation.json SHA256 dd7a8dcf4c31; full provenance in figure_evidence.json

Exact distribution of valid FM_P configurations

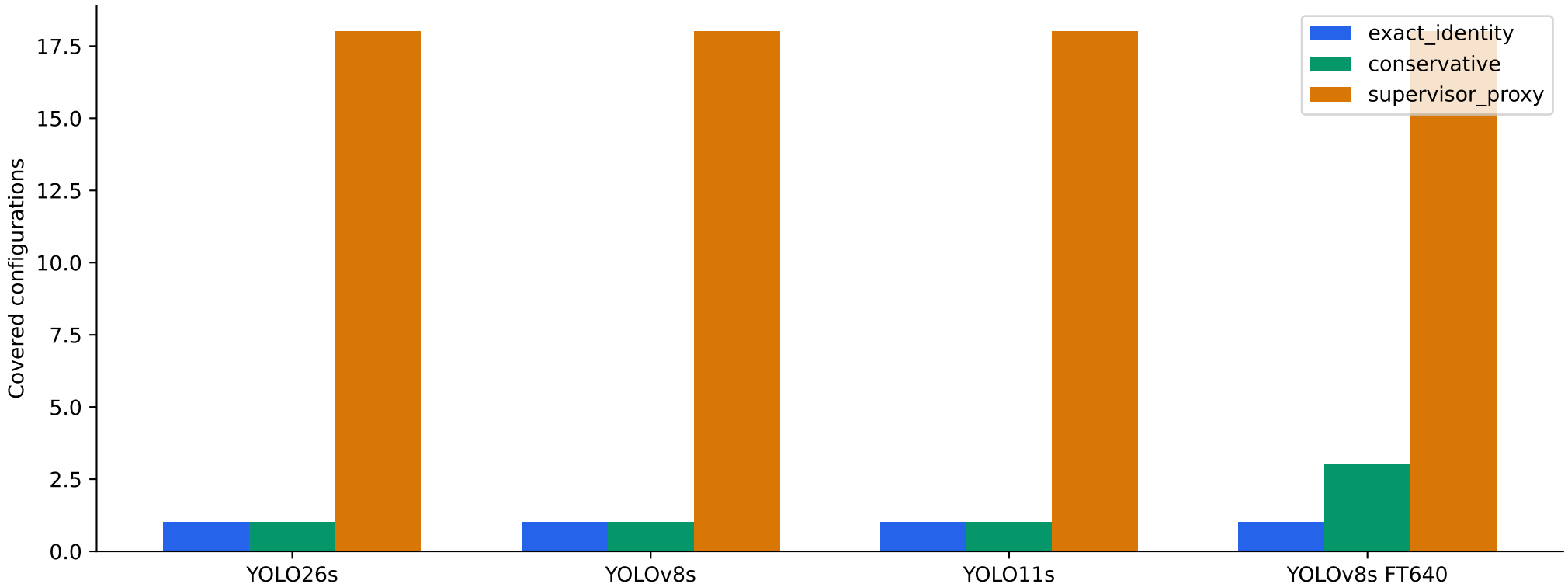


EXACT_ENUMERATION | Units: configuration count

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: configuration_distribution.csv SHA256 4d0b4b96742f; full provenance in figure_evidence.json

Static coverage depends on the evidence policy

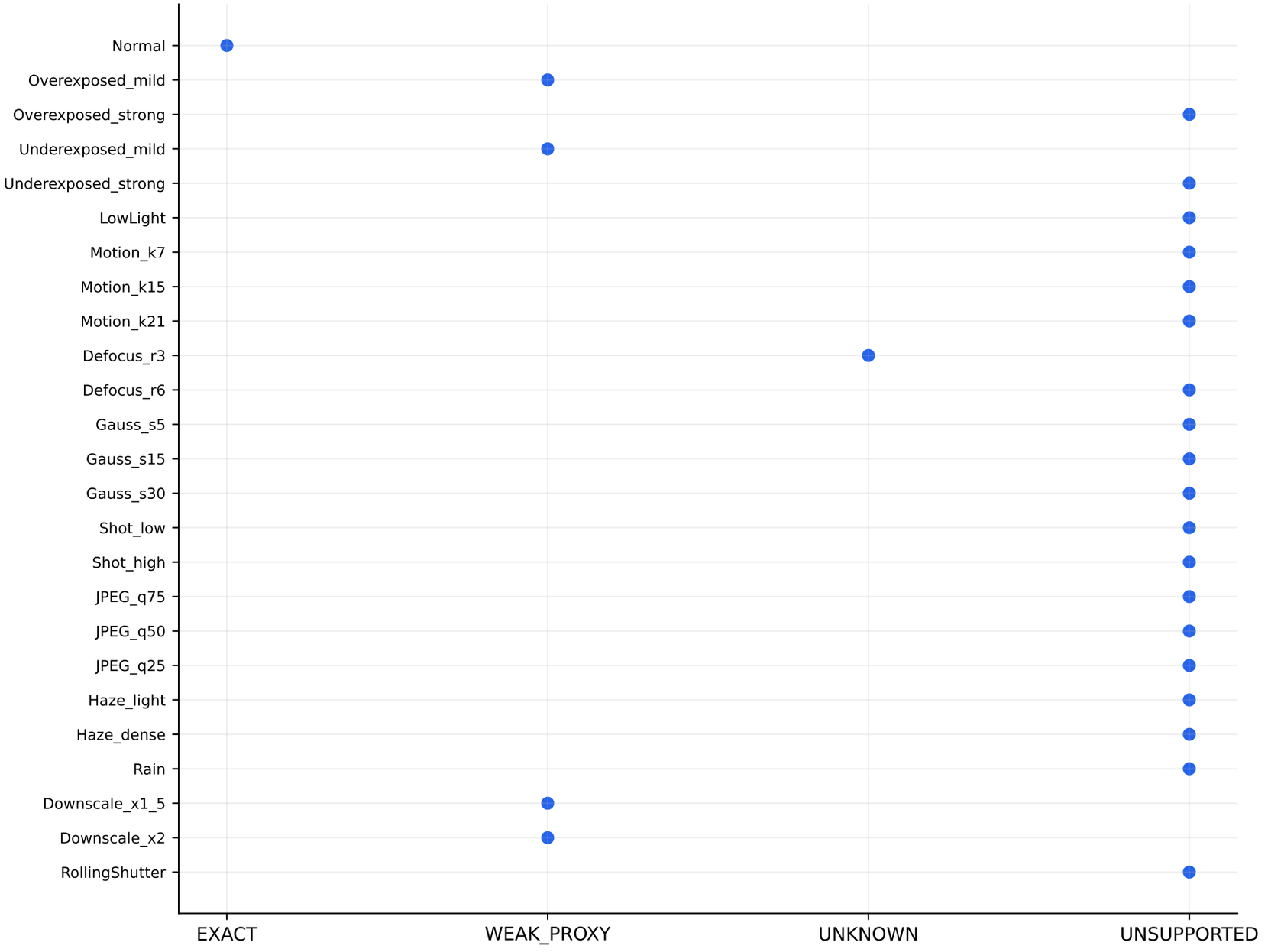


STATIC_ONLY; not detector failure | Units: covered FM_P configurations out of 46,368

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: static_coverage_by_checkpoint.csv SHA256 ef07d61fe9d3; full provenance in figure_evidence.json

Training-source coverage by degradation feature

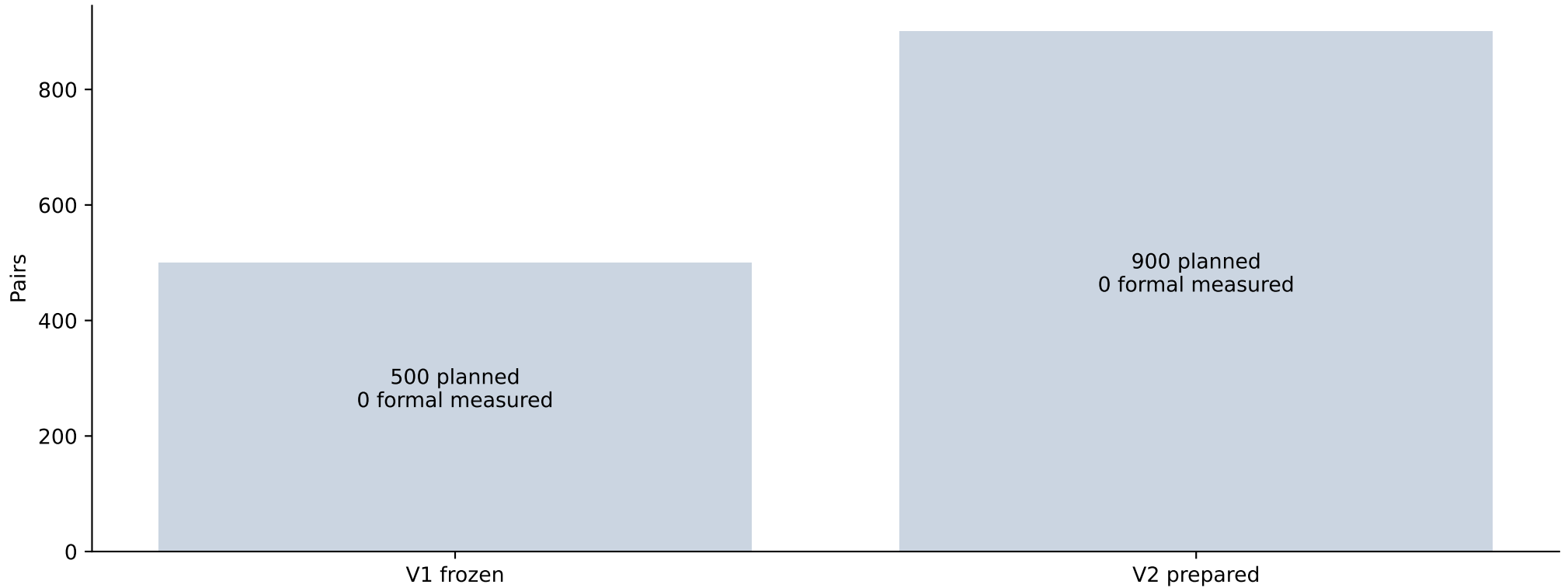


CV261102_SOURCE_POLICY; historical activation limits retained | Units: classification, not AP

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: static_baseline.json SHA256 b5cbacc3b975; full provenance in figure_evidence.json

Sampling design and formal execution coverage

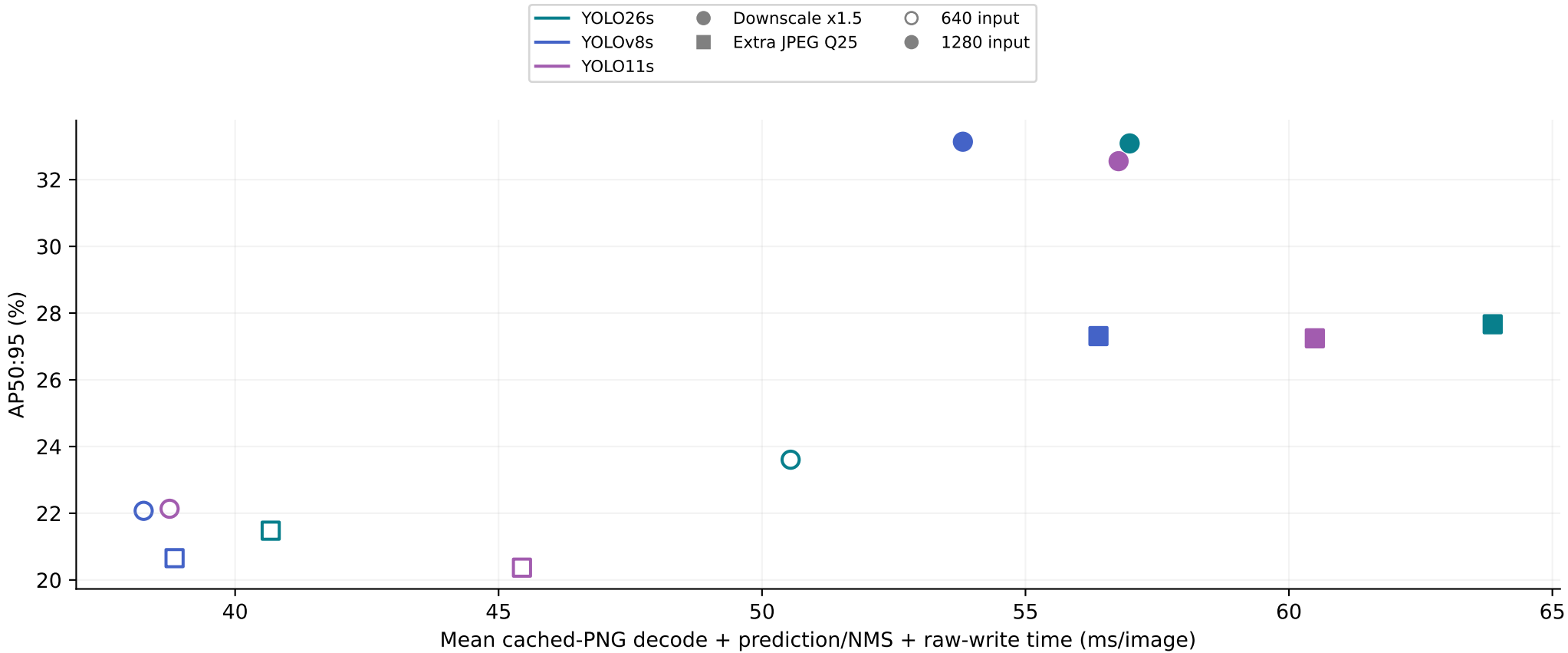


PLANNED_ONLY; DATASET_BLOCKED; no formal pairs | Units: pair count

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: cross_architecture_v2.json SHA256 e06b7a40d0d6; full provenance in figure_evidence.json

Measured AP and cached-input pipeline cost



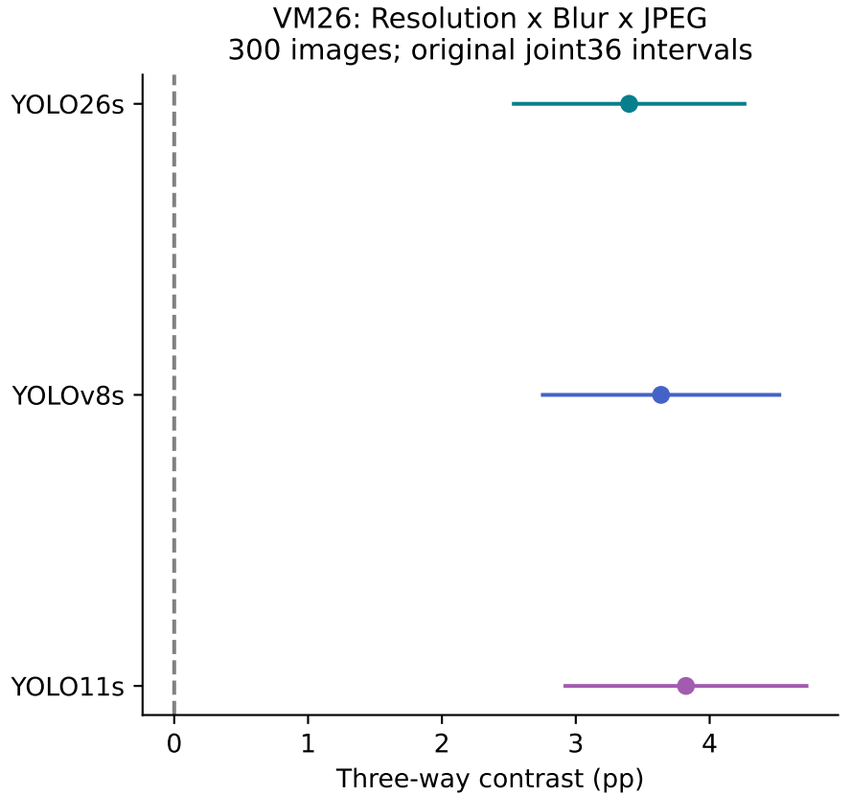
Generation, cold initialization and GPU occupancy are separate from the displayed pipeline.

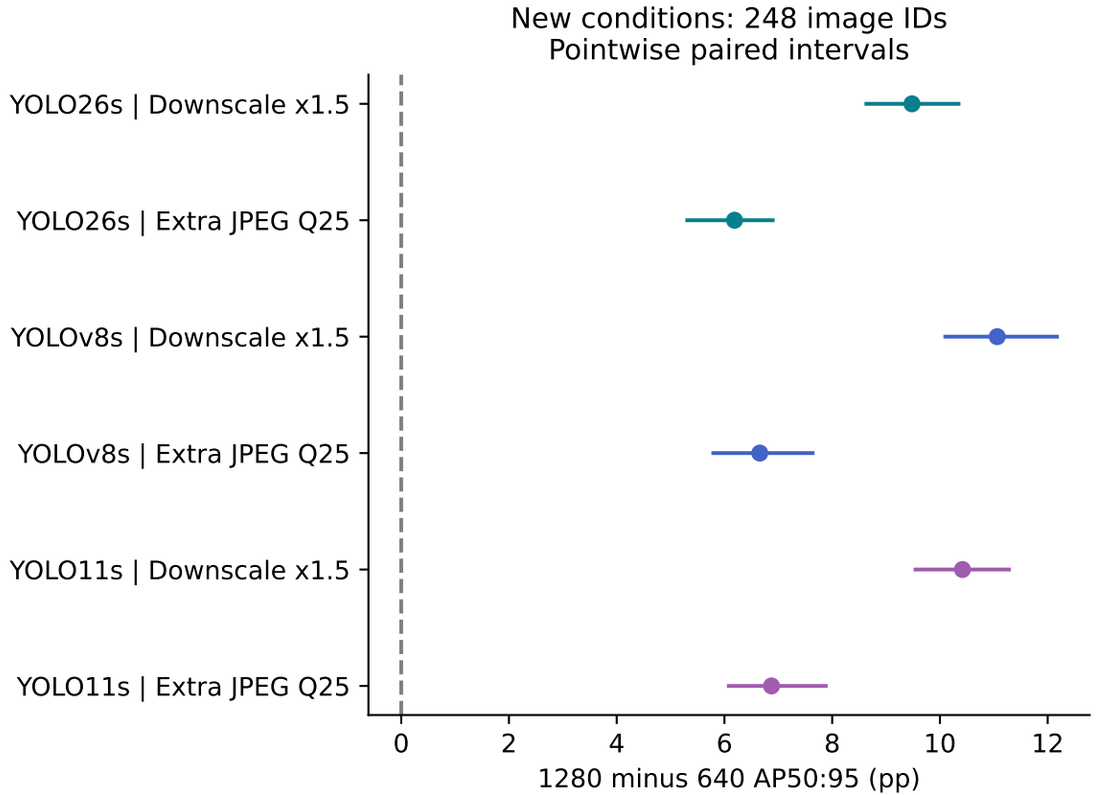
DEVELOPMENT_ONLY | 248 image IDs / 247 distinct image byte hashes | native10 | FP32

Pointwise image bootstrap; no independent test or training-seed inference.

cv261207_development_analysis_20261108T194456_6e430eb3 | source configuration_metrics.csv SHA a6dc72e0008d

Historical interaction and new same-checkpoint contrasts





DEVELOPMENT_ONLY | Left: historical 300; right: new 248 image IDs | Separate cohorts; AP is not pooled.

Left: original simultaneous 36-contrast family. Right: pointwise paired intervals. Neither is independent-test evidence.

cv261207_development_analysis_20261108T194456_6e430eb3 | source resolution_effects.csv SHA 9a75093c3f3f

Measured subset: Tiny AP >= 0.10 and mean pipeline <= 100 ms

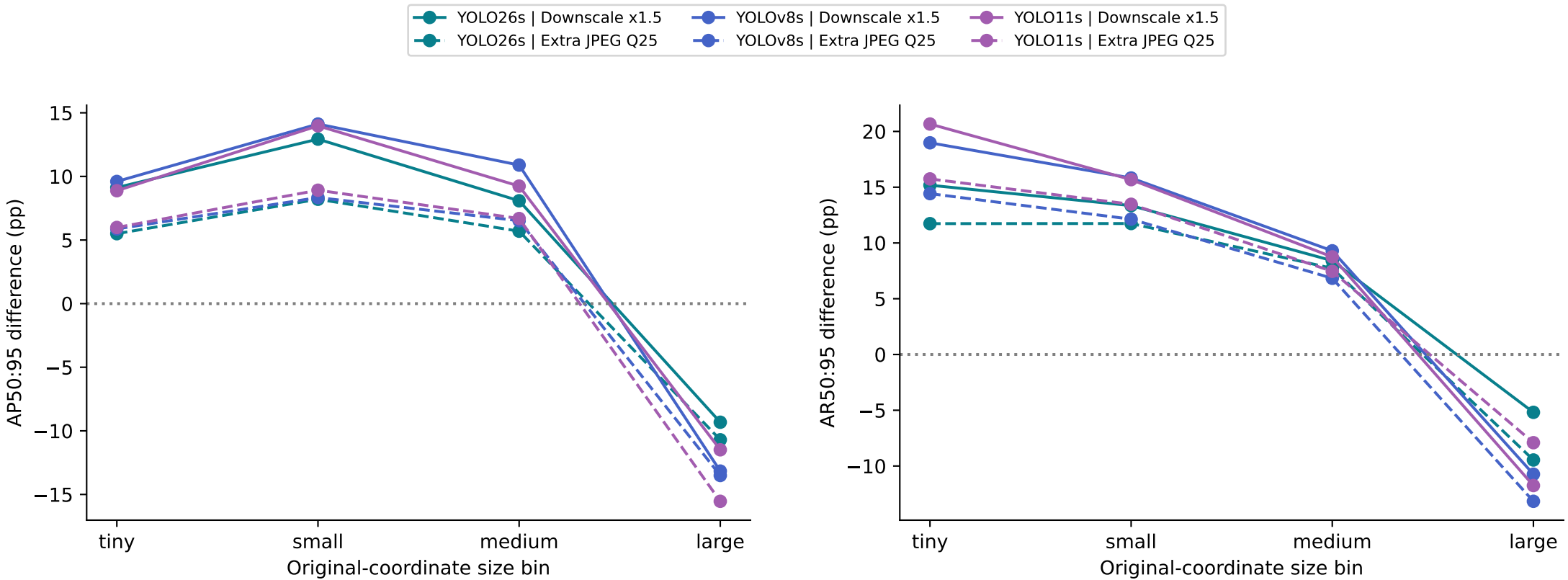


DEVELOPMENT_ONLY | 248 image IDs / 247 distinct image byte hashes | native10 | FP32

Pointwise image bootstrap; no independent test or training-seed inference.

cv261207_development_analysis_20261108T194456_6e430eb3 | source pair_states.csv SHA 79f5edb21f4b

Size-dependent resolution effects: AP and AR



1280 minus 640, same checkpoint. Point estimates shown; paired intervals are in the source table.

DEVELOPMENT_ONLY | 248 image IDs / 247 distinct image byte hashes | native10 | FP32

Pointwise image bootstrap; no independent test or training-seed inference.

cv261207_development_analysis_20261108T194456_6e430eb3 | source resolution_effects.csv SHA 9a75093c3f3f

RQ4 engineering readiness; efficacy remains unmeasured

Clean-control canary: CANARY_PASS

Actual network exposures: 64

Actual optimizer updates: 1

Targeted augmentation arms: NOT_EVALUATED

Matched before/after gap closure: NOT_EVALUATED

No efficacy curve is available. No long training was started.